Towards Intelligent Defence Surveillance: A Systematic Review of Multi-Modal Sensor Fusion, Multi-Agent AI, and Real-Time Threat Classification Frameworks

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***Abstract*—India's Defence and critical-infrastructure surveillance network spans thousands of kilometers, with critical information scattered across unconnected sensors and surveillance systems. This fragmentation, along with human operation for threat detection and classification, makes the system suffer from high false-alarm rates, which constrain timely and reliable threat assessment. Recent advancements in technology, AI, and Machine Learning offer new methods to enhance the system, such as multi-sensor fusion and AI analysis for automating surveillance systems, leaving only the critical decision-making part with manual operators. This survey provides a structured review of the five major domains of such an AI-based system that can potentially automate surveillance and reduce operators' fatigue and high false-alarm rates, providing accurate threat detection and classification. Our principal observation is that despite the growth in AI and ML, there exists no such framework that unifies diverse sensors and performs accurate threat detection and five-level threat classification across multiple domains. These gaps motivate the design and study of a unified, multi-agent, AI-driven surveillance platform tailored to Indian Defence requirements.**

I. Introduction

India maintains active surveillance obligations along more than 15,000 km of land borders with Pakistan, Bangladesh, Myanmar, and Nepal, a 7,500 km coastline, and an expanding portfolio of critical infrastructure installations [1]. Each segment employs its own constellation of radars, thermal imagers, unattended ground sensors, electro-optical cameras, and communication relays, generating data volumes that far exceed the analytical capacity of human operators [2]. Although individual sensor modalities have matured considerably, the absence of a unified reasoning layer that correlates signals across modalities and geographic sectors leaves the surveillance enterprise structurally brittle. A scientometric analysis of defence-technology publications between 2015 and 2025 confirms a fourfold increase in AI-related land-combat research, yet integrated multi-domain surveillance architectures remain conspicuously under-represented [3].

Two compounding failures intensify this structural weakness. First, false-alarm rates across deployed perimeter-security systems impose a cumulative attentional tax on operators: every spurious trigger diverts cognitive and physical resources from genuine threats [4]. Second, existing alarm architectures lack graduated threat classification; a perimeter sensor trip caused by wildlife and a coordinated tactical incursion receive equivalent system-level urgency, preventing prioritised resource allocation [5]. Decision-support systems that incorporate explainability mechanisms such as SHAP or attention-based saliency maps have shown promise for anomaly triage in civilian surveillance [6], but their adaptation to military operational tempo and accountability requirements remains an open problem [7].

Multi-agent AI architectures offer a principled response to both failures. Rather than replacing human judgment, a network of specialised software agents can function as a filtering and escalation layer that operationalises algorithmic situational awareness: each agent ingests a defined sensor stream, applies domain-specific inference, and feeds a threat-correlation agent that synthesises inputs into a graduated alert [8], [9]. Such architectures have demonstrated measurable gains in cyber-threat correlation and battlefield command support [10], and purpose-built human-autonomy teaming frameworks preserve commander authority over critical engagement decisions [11], [12]. Extending this paradigm to physical-domain defence surveillance requires integration across five technical pillars that, to date, have been investigated largely in isolation.

This paper makes two contributions. First, it provides a structured literature review spanning multi-modal sensor fusion [2], multi-agent threat detection [8], [9], false-alarm reduction and explainable AI [4], [6], geospatial awareness with human-in-the-loop control [13], [14], and real-time distributed architectures with adversarial robustness [15], [16]. Second, it identifies and formalises the integration gap across these five domains as the primary bottleneck obstructing India's transition from sensor-rich but intelligence-poor surveillance to a unified, graduated threat-management grid. Section II traces the historical and policy background. Sections III through VII examine each domain. Section VIII presents a comparative taxonomy, analyses open gaps, outlines a proposed architectural direction, and concludes the paper.

II. Background

India's post-independence border-management doctrine evolved from a manpower-centric model predicated on physical fencing, foot patrols, optical observation posts, and high-frequency radio communication [1]. For decades, situational awareness at the tactical edge depended almost entirely on the sensory acuity of deployed personnel, supplemented by rudimentary surveillance aids such as binoculars, trip-wire alarms, and vehicle-mounted searchlights. This paradigm, while operationally familiar, scaled poorly against the geographic enormity and terrain diversity of India's borders and offered minimal capability for persistent, all-weather monitoring.

The Comprehensive Integrated Border Management System (CIBMS) represents the most significant doctrinal and technological inflection point in Indian border surveillance. Conceived as a replacement for the purely physical barrier model along the India–Pakistan and India–Bangladesh frontiers, CIBMS aggregates thermal imagers, infrared and laser-based intruder-detection alarms, unattended ground sensors, aerostat-mounted radars, underwater sonar arrays, and fibre-optic fence-disturbance sensors into a networked command-and-control (C2) architecture [1]. Subsequent augmentations have introduced battlefield and human-detection radars along contested sectors, as well as seismic sensors designed to detect cross-border tunnelling activity. While CIBMS marked a decisive shift from isolated sensing toward integrated C2, the system's alert logic remains predominantly rule-based and threshold-driven, generating alarms that lack contextual discrimination [5].

A parallel transformation has occurred in the global military-surveillance research community, where rule-based analytical methods have given way to machine-learning and deep-learning pipelines capable of processing sensor data at volumes and velocities that exceed human cognitive throughput [17]. Convolutional architectures such as YOLOv8 have demonstrated strong detection performance for military vehicles, weapon systems [18], and military assets in satellite imagery [19]. Temporal models combining CNNs with LSTMs and more recent transformer-based designs have improved anomaly detection in continuous video streams while reducing false-positive rates [4]. The U.S. Department of Defense's Project Maven programme, which employed multi-vendor AI pipelines for full-motion video exploitation, provided an early operational proof that a 20-person AI-augmented team could match the throughput of a 2,000-person manual targeting cell [20], [21].

India's defence establishment has responded with a series of policy signals and institutional investments. The Ministry of Defence has identified artificial intelligence as a priority technology area and expanded the Defence Research and Development Organisation's (DRDO) Centre for Artificial Intelligence and Robotics (CAIR) mandate to include autonomous surveillance, target recognition, and AI-enabled command systems [1]. Commercial AI vendors and data-fusion platforms have been evaluated for cross-database correlation and anomaly detection in intelligence workflows [22]. Nonetheless, analysis of India's defence-AI portfolio reveals a pattern of demonstration-grade pilots that remain stovepiped within individual service branches, lacking the interoperability and shared ontological standards necessary for cross-domain fusion [1].

Several persistent challenges motivate the present survey. Bandwidth constraints in remote and contested border zones limit the throughput of cloud-centric inference pipelines, necessitating edge-computing architectures that can perform preliminary threat assessment at the sensor node [15]. Environmental noise sources, from sandstorms along the Rajasthan frontier to monsoon-driven river-level fluctuations along the Bangladesh border, inject sustained false-alarm stimuli that overwhelm threshold-based detection logic [4]. The opacity of deep-learning classifiers raises accountability concerns for military decision-makers who must justify escalatory actions under international humanitarian law [7], [12]. Legacy sensor systems deployed across CIBMS sectors lack standardised data interfaces, complicating real-time fusion with newer AI-enabled platforms [2]. Adversarial actors, meanwhile, are increasingly capable of exploiting vulnerabilities in deployed object-detection models through physical-world adversarial patches and GPS spoofing [16], [23]. Taken together, these unresolved challenges define the integration gap that this survey examines, beginning with a domain-by-domain taxonomy of the five constituent technical pillars.

III. Taxonomy of Related Work

The surveyed literature was partitioned into five functional domains that collectively span the processing pipeline of an integrated defence surveillance system: (A) multi-modal sensor fusion and data integration, (B) multi-agent AI for threat detection and classification, (C) false-alarm reduction and explainable decision support, (D) geospatial situational awareness and human-in-the-loop governance, and (E) real-time distributed architecture and adversarial robustness. Inclusion criteria restricted the corpus to studies satisfying at least two of three requirements: applicability to defence or security-critical environments, employment of AI or machine-learning methods, and involvement of multi-sensor or distributed system architectures. Sources span peer-reviewed journal articles, IEEE and ACM conference proceedings, and policy-research technical reports [20], [22].

Table I presents the resulting classification. For each domain the table identifies representative published approaches, the principal sensor or data modalities examined, the key technique employed, and the primary limitation observed when the approach is evaluated against the requirements of a unified multi-domain surveillance framework. The five domains are not mutually exclusive in operational practice: fusion quality at the sensor-integration layer (Domain A) directly conditions the classification confidence achievable by downstream agents (Domain B), and explanation-generation latency in the decision-support layer (Domain C) constrains the responsiveness of human authorisation loops (Domain D).

TABLE I
TAXONOMY OF DEFENCE SURVEILLANCE LITERATURE

Domain	Representative Approaches	Sensor / Data Modality	Key Technique	Limitation Noted
A: Sensor Fusion	Multi-source information fusion for C3I systems; attention-based cross-modal alignment	Radar, thermal, acoustic, optical, RF	Feature-level and decision-level multi-source fusion	Evaluated on pairwise modalities only; no adversarial degradation testing
B: Multi-Agent AI	Modular MAS with role-specialised sub-agents; YOLOv8 military object detection	Video, radar returns, satellite imagery, system logs	Agent decomposition with cross-source threat correlation	Homogeneous task distribution; no cross-modal agent orchestration
C: False Alarm / XAI	Deep-learning anomaly detection; SHAP/LIME explanation methods	Surveillance video, CCTV feeds, IDS telemetry	Temporal consistency filtering; post-hoc feature attribution	XAI latency not evaluated within real-time alert pipelines
D: Geospatial / HITL	GIS-based threat dashboards; trust calibration for autonomy teaming	GIS terrain layers, sensor metadata, force positions	Terrain-aware threat mapping; human authorisation loops	Tested in single-operator, single-threat scenarios only
E: Distributed / Adversarial	Kafka + Flink real-time streaming; adversarial patch attacks on YOLO detectors	IoT telemetry, edge video, RF uplinks	Edge preprocessing with stream-processing inference; adversarial training	No evaluation under simultaneous node failure and adversarial injection

These cross-domain dependencies are analysed quantitatively in Section V.

IV. Literature Survey

A. Multi-Modal Sensor Fusion and Data Integration

1) Problem Definition: Defence surveillance sensors operate across incompatible modalities. Satellite imagery provides wide-area coverage at low temporal frequency; unmanned aerial vehicle (UAV) feeds deliver narrow-area, high-frame-rate video; ground-based sensors including seismic, acoustic, and infrared detectors produce numeric time-series data with distinct noise profiles and sampling rates [2]. Reconciling these heterogeneous data formats, sampling cadences, transmission latencies, and reliability levels into a coherent, low-latency operational picture constitutes the central technical challenge of sensor fusion for border security. Terrain variability across Indian border sectors, from arid desert to dense riverine mangrove, and intermittent radio and satellite connectivity compound the integration problem considerably.

2) Existing Approaches: The Joint Directors of Laboratories (JDL) data fusion model defines five processing levels from signal-level refinement (Level 0) through situation assessment (Level 2) to threat refinement (Level 3) and remains the dominant architectural reference for military fusion systems [2]. A comprehensive survey of multi-source information fusion for command, control, communications, and intelligence (C3I) systems by Meng et al. [2] distinguishes data-level, feature-level, and decision-level fusion layers, cataloguing the computational tradeoffs inherent in each. Attention-based cross-modal feature alignment has yielded the highest classification accuracy in counter-UAV applications when individual sensor reliability varies across optical, thermal, radio-frequency, and radar inputs. At the industry level, defence contractors have integrated hyperspectral imaging with AI-enabled radar and LiDAR processing for military situational awareness, and passive infrared sensor networks fused with multi-modal inputs have been deployed for autonomous detect-classify-track operations at military installations. Ali et al. [17] survey transformer-based models and federated learning pipelines for real-time threat detection, identifying attention mechanisms as particularly effective for cross-modal feature alignment.

3) Key Insight and Research Gap: Reviewed fusion architectures are predominantly evaluated on homogeneous or pairwise sensor combinations and have not been validated against the five-or-more modality configurations present in deployed systems such as the Comprehensive Integrated Border Management System (CIBMS). Temporal misalignment between high-frequency ground sensors reporting at 100 Hz or above and low-frequency satellite revisit passes separated by hours remains an unsolved synchronisation problem [2]. No study in the reviewed corpus evaluates fusion performance under adversarial sensor degradation, a gap that carries direct operational significance given documented electromagnetic interference threats in contested border environments [23].

B. Multi-Agent AI for Threat Detection and Classification

1) Problem Definition: A single monolithic inference model cannot simultaneously perform video-based object detection across multiple camera streams, radar-based anomaly classification, ground-sensor time-series analysis, and cross-source threat correlation at the sub-second latencies demanded by operational response protocols [10]. Distributing these functions across specialised software agents introduces coordination requirements: agents must share intermediate inferences, resolve conflicting threat assessments originating from disparate modalities, and converge on a unified threat classification without creating a centralised processing bottleneck [8].

2) Existing Approaches: Hassan et al. [8] propose a modular multi-agent system (MAS) for cybersecurity threat detection in which specialised sub-agents, a log analyser, an IP analyser, and a threat correlator, operate independently and contribute assessments to a central correlation engine; role-based decomposition is shown to improve both detection granularity and diagnostic traceability relative to monolithic alternatives. A cooperative AI staff system managing over 500,000 simulated battlefield agents was described by Zhao et al. [10], providing command-level situational awareness through automated spatio-temporal pattern correlation. The AgenticCyber framework [9] extends multi-agent orchestration to multi-modal security domains by fusing cloud logs, surveillance video, and audio streams, achieving a reported 96.2% F1 score at 420 ms end-to-end latency. Within the visual detection component, YOLOv8-family architectures have been adapted to military targets across satellite imagery [19] and weapon identification in live surveillance feeds [18]. Project Maven's operational deployment demonstrated that a 20-person AI-augmented cell could match the throughput of a 2,000-person manual targeting team [20], although independent evaluation reported only 60% tank identification accuracy under favourable conditions [21]. Human-autonomy teaming frameworks that partition decision authority between AI agents and human commanders have been formalised for tactical operations [11].

3) Key Insight and Research Gap: Existing MAS surveillance architectures address homogeneous task distributions, such as multi-camera handoff across a uniform video network, and do not tackle heterogeneous cross-modal agent orchestration in which agents reason over fundamentally different data representations including pixel arrays, spectrograms, and radar range-Doppler maps [8], [9]. A graduated five-level threat classification scheme requires each agent to produce calibrated confidence scores and the correlation agent to perform principled uncertainty quantification when fusing outputs from agents with disparate error profiles. Current MAS designs treat confidence calibration as an afterthought rather than a first-class architectural constraint [22].

C. False-Alarm Reduction and Explainable Decision Support

1) Problem Definition: Persistently high false-alarm rates constitute a documented and operationally consequential failure mode in fielded surveillance systems [4]. Sustained spurious alerts induce operator alert fatigue, misallocate rapid-response resources, and progressively degrade trust in automated detection pipelines. Root causes include environmental noise from wind-induced fence vibration, animal crossings, and weather-driven radar clutter; absence of temporal consistency verification in threshold-based alarm logic; and lack of contextual priors that distinguish known civilian activity zones from restricted perimeters [5].

2) Existing Approaches: Weakly-supervised multiple-instance learning (MIL) frameworks for anomaly detection in real-world surveillance video have established training paradigms that substantially reduce annotation requirements for large-scale deployment [4]. Deep-learning surveys catalogue CNN, LSTM, and transformer-based temporal models that improve anomaly discrimination by enforcing signature persistence across consecutive frames before triggering an alert. Hybrid architectures combining spatial object detection with temporal verification have been shown to reduce false-positive rates relative to single-frame classifiers in CCTV monitoring [5]. On the explainability front, Zhang et al. [6] survey SHAP, LIME, and attention-based explanation mechanisms applicable to black-box classifiers, applying explainable AI (XAI) to intrusion detection and malware classification, tasks that are structurally analogous to the threat-triage function required in military surveillance. Roff and Danks [7] address military-specific accountability requirements, stipulating that commanders must be able to audit the rationale behind targeting and escalatory recommendations under both domestic and international legal obligations.

3) Key Insight and Research Gap: XAI methods in surveillance have been evaluated primarily for post-hoc interpretability on static or pre-recorded datasets [6] and have not been integrated into real-time decision pipelines where explanation generation latency itself constrains operator response time. Contextual filtering approaches that suppress alarms in geofenced civilian zones depend on accurate, current civilian activity maps that are unavailable or classified in active border areas [5], creating an unresolved data-dependency bottleneck that no reviewed study has addressed.

D. Geospatial Situational Awareness and Human-in-the-Loop Systems

1) Problem Definition: Actionable command decisions require threat intelligence to be spatially anchored: operators must perceive not only the nature and severity of a detected anomaly but also its precise location relative to terrain features, installation boundaries, sensor coverage envelopes, and proximity to civilian population centres [13]. Graduated threat classification further demands that alerts assessed at the highest severity levels do not trigger autonomous system action; human authorisation must remain mandatory at those tiers [12]. The interface design challenge is twofold: preventing operators from reflexively endorsing AI recommendations while preserving response latencies compatible with tactical decision cycles [14].

2) Existing Approaches: Karimipour and Derakhshan [13] describe AI-enabled situational awareness dashboards for distributed sensor networks in industrial Internet of Things (IoT) environments, incorporating geographic information system (GIS) threat mapping with sensor registries that visualise coverage zones, detection radii, and active alarm sources on unified terrain layers. Zhao et al. [10] extend this concept to military command by constructing multi-layer spatio-temporal battlefield visualisations that overlay force positions, predicted adversary trajectories, and terrain accessibility indices. Geospatial UAV detection with movement-vector plotting on georeferenced maps has been demonstrated for counter-drone applications. Maas [14] examines trust calibration and automation bias in military human-autonomy teaming, reporting that operator trust is sensitive to the transparency and perceived reliability of AI outputs. Svenmarck and Luotsinen [12] formalise the legal and technical requirements for meaningful human control, proposing design constraints that ensure commander authority over engagement decisions is not diluted by automated recommendation systems.

3) Key Insight and Research Gap: Human-in-the-loop (HITL) frameworks in the surveyed literature focus on single-operator, single-threat interaction models [12], [14]. Concurrent multi-threat environments, in which an operator must triage AI-generated alerts across geographically distributed incidents simultaneously, remain under-examined despite constituting the operational norm in border-surveillance command centres. Interface designs that communicate prediction uncertainty quantitatively to operators with varying levels of technical expertise present an open challenge; existing prototypes either display raw probability values that non-technical operators misinterpret or suppress uncertainty information entirely, promoting uncalibrated trust [14].

E. Real-Time Distributed Architecture and Adversarial Robustness

1) Problem Definition: Sensor data generated at a remote border outpost must traverse constrained communication links, undergo inference at sub-second latency, and arrive at a central command dashboard with actionable fidelity [15]. Meeting this requirement demands edge preprocessing to reduce transmission volume, stream-processing middleware for sequential low-latency inference, and an architecture tolerant of node failures and intermittent connectivity. Adversarial robustness introduces a distinct threat surface: a high-value surveillance system constitutes an explicit target for adversarial patches designed to defeat object detectors, GPS spoofing aimed at unmanned platform navigation, and signal jamming directed at ground-sensor communication uplinks [16], [23].

2) Existing Approaches: Patel et al. [15] present a real-time IoT sensor streaming architecture combining Apache Kafka for telemetry ingestion with Apache Flink for stream processing, achieving low-latency anomaly detection on continuous sensor feeds with fault-tolerant horizontal scaling. Edge-computing extensions of this pattern perform preliminary inference at the sensor node before transmitting compressed results, reducing bandwidth requirements for drone and CCTV data in remote deployments. Ali et al. [17] survey transformer-based models and federated learning pipelines for real-time threat detection across IoT and 5G networks, identifying federated architectures as particularly suitable for privacy-preserving distributed inference in military contexts. On the adversarial front, Liang et al. [16] catalogue physical-world attacks on YOLO-family detectors, including adversarial patches and printed perturbations that cause vehicle misclassification in outdoor conditions. Wang et al. [23] address military-specific adversarial risks such as electromagnetic interference and GPS spoofing, and propose ethical governance guardrails for deployed military AI. Scientometric analysis of approximately 3,500 Scopus records confirms a fourfold growth in AI-related defence publications between 2015 and 2025 [3], yet distributed surveillance architectures remain under-represented relative to single-algorithm detection studies. Independent PRISMA reviews of UAV cybersecurity [24] and UAV intrusion detection systems [25] converge on a shared conclusion: no unified cross-domain framework currently addresses the full defence surveillance pipeline from edge to command centre.

3) Key Insight and Research Gap: Adversarial robustness research in computer vision predominantly addresses digital perturbations on laboratory-curated datasets; physical-world attacks under outdoor surveillance conditions involving variable lighting, partial occlusion, and ambient sensor noise are substantially less investigated [16]. No reviewed distributed-architecture study evaluates end-to-end pipeline latency under simultaneous edge-node failure and adversarial sensor injection, the compound threat model most representative of a contested border environment. The gap between adversarial machine-learning research and operational distributed systems engineering constitutes one of the least-addressed intersections in the defence surveillance literature [23], [24].

V. Comparative Analysis

Cross-domain evaluation reveals an uneven maturity gradient across the five surveyed areas. Sensor fusion (Domain A) and visual object detection (a component of Domain B) represent the most technically mature pillars, and YOLOv8-family detectors [18], [19] approach deployment readiness for constrained single-modality scenarios. False-alarm reduction techniques (Domain C) yield measurable improvements on benchmark datasets [4] but lack validated integration with real-time explainability pipelines and with the accountability demands specific to military decision chains [7]. Geospatial HITL systems (Domain D) and adversarially hardened distributed architectures (Domain E) remain at the proof-of-concept stage, with empirical validation limited to single-site deployments or simulated environments [13], [15].

The most consequential pattern emerging from this review is the near-total absence of research addressing interaction effects across domain boundaries. Fusion fidelity at the sensor-integration layer directly determines the confidence bounds available to downstream classification agents [2], [8]. Explanation-generation overhead in the XAI layer imposes an additive latency penalty on the HITL authorisation cycle [6], [14]. Adversarial corruption at edge sensors propagates cascading failures through fusion, classification, and alerting stages that no surveyed system models as an integrated failure mode [16], [23]. Table II summarises the best-performing technique, reported benchmark, metric, and principal limitation for each domain when assessed against the requirements of a unified multi-domain surveillance framework.

TABLE II
COMPARATIVE ANALYSIS OF SURVEYED APPROACHES

Domain	Best Performing Technique	Dataset / Benchmark	Metric Reported	Limitation in Unified Context
A: Sensor Fusion	Attention-based cross-modal DL fusion	Counter-UAV optical, thermal, RF, radar corpus	Highest classification accuracy vs. single-modal baselines	No temporal synchronisation for >2 modalities at disparate sampling rates
B: Multi-Agent AI	Multi-modal agentic orchestration	Cloud logs + surveillance video + audio	96.2% F1; 420 ms end-to-end latency	Agent confidence calibration treated as post-hoc addition
C: False Alarm / XAI	Deep-learning anomaly detection with temporal verification	Real-world surveillance video benchmarks	Benchmark AUC across anomaly classes	Explanation latency not measured for real-time deployment
D: Geospatial / HITL	Multi-layer battlefield visualisation	500K+ simulated battlefield agents	Command-level situational awareness coverage	Multi-threat concurrent operator triage not evaluated
E: Distributed / Adversarial	Kafka + Flink real-time streaming	Continuous IoT sensor streams	Low-latency, fault-tolerant stream processing	Not tested under adversarial sensor injection or node compromise

VI. RESEARCH GAP

A. Limitations Across Existing Studies

The dominant fusion and detection studies evaluated across Domain A and Domain B operate on one or two sensor modalities simultaneously. No reviewed work addresses synchronisation fidelity when five or more heterogeneous modalities, including radar, thermal imaging, seismic sensors, acoustic arrays, and satellite imagery, contribute to a shared fusion graph at incompatible sampling rates. Multi-source fusion surveys catalogue the computational tradeoffs of data-level, feature-level, and decision-level integration [2], yet treat modality count as an extensible parameter rather than a source of combinatorial complexity in temporal alignment and confidence propagation. Pairwise evaluation protocols create an illusory impression of scalability that collapses once the modality count of a deployed border system such as CIBMS is approached.

A second cross-cutting limitation concerns classification granularity and confidence calibration. Multi-agent and anomaly-detection architectures reviewed in Sections IV-B and IV-C produce binary or coarse-class outputs: threat versus non-threat, or anomalous versus normal [8], [9]. None implements a graduated five-level classification scheme with per-level confidence calibration, formal uncertainty quantification, and documented decision boundaries separating adjacent threat tiers. This is not solely an engineering deficit; it is an operational one. Without calibrated confidence at each severity level, human operators possess no rational basis for proportional resource allocation, and response protocols collapse to a single default escalation posture regardless of actual threat magnitude.

Explainability methods surveyed in Domain C exhibit a latency profile that is structurally incompatible with the HITL throughput requirements identified in Domain D. SHAP and LIME generate explanation artefacts through iterative perturbation or kernel-based approximation at timescales ranging from hundreds of milliseconds to several seconds per input instance [6]. Military accountability frameworks stipulate that decision-makers must have access to auditable rationale before authorising escalatory action [7], yet the alert pipelines into which these explanations must be inserted demand sub-second end-to-end delivery [14]. When explanation latency exceeds the operator's decision window, the HITL layer ceases to function as a deliberative safeguard and becomes instead a throughput bottleneck that delays response without improving decision quality.

B. Missing Elements in Current Research

No reviewed system models the compound failure mode in which adversarial sensor corruption at the network edge propagates through fusion, classification, explanation, and alerting stages within a single end-to-end evaluation. Physical-world adversarial attacks on object detectors are investigated in isolation on clean benchmarks [16], and military AI risk assessments catalogue GPS spoofing, electromagnetic jamming, and data poisoning as independent threat vectors [23]. The distinction between digital-perturbation adversarial research, conducted on curated datasets under controlled illumination, and the physical operational reality of contested Indian border environments, where variable weather, terrain occlusion, and deliberate countermeasures co-occur, is neither acknowledged nor addressed in any integrated evaluation framework.

The surveyed literature also lacks publicly available benchmarks derived from Indian border terrain, sensor configurations, or threat typologies. Models trained and validated on standard surveillance benchmarks or NATO-oriented military imagery [19] carry embedded domain assumptions about scene geometry, illumination statistics, and threat-actor behaviour that do not transfer to Himalayan high-altitude posts, Thar Desert thermal environments, or the Sundarbans riverine terrain. This constitutes both a data-availability gap and a policy gap: operational sensor data from CIBMS deployments is necessarily classified, precluding open benchmarking, while India's defence-AI research community has not yet established synthetic or semi-synthetic equivalents calibrated to domestic operational conditions [1].

The five technical domains surveyed in Sections IV-A through IV-E are treated throughout the literature as independent research streams. No reviewed work evaluates a system architecture that jointly optimises sensor fusion fidelity, multi-agent coordination overhead, explanation latency, HITL authorisation throughput, and adversarial robustness as a coupled system of interacting constraints [2], [15], [25]. The integration gap formalised in Section I is therefore confirmed as structural rather than incidental: the obstacle to unified defence surveillance is not the absence of solutions to individual sub-problems but the absence of a framework that treats their mutual interactions as first-class design constraints.

VII. Proposed Direction

The research gaps identified above motivate an architectural direction organised around four tightly coupled subsystems. The first subsystem is a hierarchical multi-modal fusion backbone in which edge inference nodes perform modality-local preprocessing and lightweight anomaly scoring at the sensor tier, transmitting compressed intermediate representations to a regional aggregation tier [15]. At the aggregation tier, learnable synchronisation buffers reconcile cross-modal temporal misalignments arising from disparate sampling rates across radar, thermal, seismic, acoustic, and satellite inputs. This architecture must be evaluated under five-or-more modality combinations and must quantify classification degradation under adversarial sensor dropout, extending established fusion models with explicit robustness metrics absent from current benchmarks [2].

The second subsystem is a role-specialised multi-agent orchestration layer in which each agent is assigned a defined input modality, inference scope, and output schema. Agents communicate through a shared blackboard with a threat-correlation arbitrator agent responsible for producing a graduated five-level classification. The arbitrator's output must comprise per-level softmax confidence scores calibrated against a held-out validation set constructed from India-relevant operational conditions [8], [9]. Uncertainty quantification, whether through Monte Carlo dropout, ensemble disagreement, or conformal prediction intervals, must be a first-class output field rather than a diagnostic afterthought, enabling downstream HITL governance to reason over the arbitrator's epistemic state [10].

The third subsystem integrates explanation generation into the alert pipeline as a constrained-latency inference component. Attention-based saliency maps and compressed SHAP approximations should be generated within the same forward pass as the threat classification, bounding explanation latency to a fixed fraction of the overall inference budget [7]. These explanations must be rendered as geospatially anchored overlays on the GIS command interface [13], presenting operators with spatially contextualised rationale rather than abstract feature-attribution vectors. Interface evaluation protocols must assess operator decision quality across expertise levels to quantify and mitigate automation bias [12], [14].

The fourth subsystem subjects the complete pipeline, from edge sensor through stream processor to command dashboard, to a compound adversarial threat model incorporating simultaneous node failure, adversarial patch injection on visual feeds, and GPS spoofing of unmanned platforms [15], [16], [23]. Human authorisation must be preserved for threat classifications at Level 4 and Level 5, with interface constraints that prevent reflexive endorsement by presenting alternative classification hypotheses and associated confidence intervals to the operator [11], [12]. End-to-end pipeline latency, classification accuracy, and explanation fidelity must all be reported under adversarial conditions as primary evaluation metrics, not supplementary ablations.

VIII. Conclusion

This survey has examined the defence surveillance literature across five technical domains: multi-modal sensor fusion, multi-agent AI for threat detection and classification, false-alarm reduction with explainable decision support, geospatial situational awareness with HITL governance, and real-time distributed architectures hardened against adversarial interference. Each domain contains mature component-level solutions that demonstrate strong performance on domain-specific benchmarks. The central finding, confirmed through cross-domain comparative analysis in Section V and formalised in Section VI, is that no existing architecture addresses the interactions among these five domains as coupled design constraints within a single operational framework. The integration gap is structural: fusion fidelity conditions classification confidence, explanation latency constrains HITL throughput, and adversarial corruption at the edge cascades through all downstream processing stages.

The architectural direction outlined in Section VII constitutes a research agenda rather than a finished system. Three open problems carry the highest priority for advancing this agenda: the construction of India-specific benchmarks calibrated to domestic terrain, sensor configurations, and threat typologies; the design and calibration of a graduated five-level threat classification scheme under realistic sensor noise and modality dropout; and the integration of bounded-latency explainability into real-time alert pipelines without degrading classification throughput. Addressing these problems in an integrated evaluation framework would represent a substantive step toward transforming India's sensor-rich but intelligence-fragmented surveillance infrastructure into a unified, graduated threat-management capability.

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